Enhancing Cognitive Digital Twin Interaction using an LLM Agent

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***Abstract*—This paper introduces a conceptual architecture design aimed at enhancing interactions with cognitive digital twins of countries through an Large Language Model (LLM) agent. By leveraging sophisticated data retrieval and summarization techniques, the architecture integrates data from diverse sources, including environmental sensors, web pages, and human inputs, to create a dynamic and comprehensive digital twin. The LLM agent facilitates intuitive conversational interfaces, allowing users to query and interact with the digital twin in a natural manner. Through advanced natural language processing and prompt engineering, the agent can understand complex queries, retrieve relevant data, and provide transparent and explainable insights. Additionally, the system incorporates a feedback loop for continuous improvement based on user interactions. This approach addresses significant challenges in data acquisition and management, offering a scalable solution for creating accurate and real-time representations of countries. The architecture aims to empower decision-makers with precise, actionable insights for policy-making, urban planning, and resource management, demonstrating a significant step towards realizing the potential of digital twins in understanding and managing complex national systems.**

Keywords*—*cognitive digital twin, LLM agent, multimodal data, real-time retrieval

I. Introduction

A digital twin is a virtual replica of a physical entity, process, or system. This concept leverages data, real-time updates, and machine learning to create a dynamic model that mirrors its physical counterpart in the digital space. By doing so, it enables detailed analysis, simulation, and prediction of performance and potential issues, ultimately enhancing decision-making and driving innovation.

Expanding on this concept, a digital twin of a country is a highly sophisticated digital model that represents the country's systems and dynamics, such as its infrastructure, environmental conditions, population, and economic systems. This comprehensive model integrates vast amounts of data from various sources, including IoT sensors, satellites, and statistical data, to simulate the country's behavior and interactions. The aim is to enable better policy-making, urban planning, and resource management by providing insights into complex national systems, predicting outcomes of various scenarios, and identifying potential areas for improvement in real-time. This tool can be pivotal in addressing national challenges like urban development, environmental sustainability, and disaster response.

Creating a digital twin of a country presents significant challenges, primarily due to the complexities in data acquisition and management. One of the primary obstacles is the lack of readily available data through structured APIs. In an ideal scenario, a digital twin would require a continuous stream of real-time, comprehensive data, which structured APIs can efficiently provide. However, the reality is that much of the necessary data is scattered across various webpages in different formats.

This disparity in data formats and sources presents a significant challenge when creating a cohesive and accurate digital model. Data can exist in unstructured forms, such as textual reports, PDFs, or embedded within complex web interfaces, which complicates the extraction and standardization process. Additionally, data accessibility and reliability levels may vary, further adding complexity. To overcome these obstacles, sophisticated data aggregation and processing techniques, including web crawling, natural language processing, and advanced data integration methods, are necessary. These approaches ensure that the digital twin faithfully represents the multifaceted nature of a country's systems and dynamics.

Our approach revolves around creating a user-friendly interface where analysts can query and interact with the digital twin in a conversational manner, similar to how they would engage in a discussion with a human expert. By leveraging advanced natural language processing (NLP) techniques, these agents can comprehend and address complex queries, translating them into actionable insights derived from the underlying digital twin model.

Furthermore, a critical aspect of our architecture is its transparency. It's designed not just to provide answers, but also to explain and rationalize those answers. This means that when an analyst queries the digital twin, the agent not only delivers the requested information or prediction but also provides an explanation of how it arrived at that conclusion. This could involve outlining the data sources used, the assumptions made, and the reasoning behind its analysis. Such transparency is essential for building trust in the system and enabling users to make informed decisions based on the insights provided. By offering explanations, the agent helps analysts understand the context and limitations of the data, leading to more nuanced and effective use of the digital twin.

Additionally, an integral feature of this automated agent

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is its ability to collect and utilize feedback from users to enhance its responses and functionality. When analysts interact with the digital twin via the agent, they have the opportunity to provide feedback on the accuracy and relevance of the information and explanations provided. This feedback mechanism is crucial for several reasons.

Firstly, it allows for the correction and refinement of the agent's responses. If an analyst finds an answer lacking in accuracy or detail, their feedback can prompt the agent to adjust its data processing or reasoning methods. Secondly, user feedback can highlight the need to incorporate new and relevant data sources into the digital twin. Analysts, with their domain expertise, are often aware of emerging data streams or underutilized sources that could enhance the model's accuracy and comprehensiveness. By signaling these resources through feedback, they can guide the development team to integrate these sources into the digital twin's data ecosystem.

This feedback loop fosters a dynamic and evolving interaction between users and the digital twin. It ensures that the agent remains responsive to the changing needs and complexities of the environments it models, thereby maintaining its relevance and efficacy as a decision-support tool.

II. Related Work

This section provides an overview of the existing research in the domain of digital twins, focusing particularly on the integration and interaction of LLMs with these digital counterparts.

A. Digital twins

Building on the transformative concept of Digital Twins, which create virtual models to mirror physical objects or systems for in-depth analysis, prediction, and management throughout their lifecycle, Cognitive Digital Twins (CDTs) represent a further evolution. CDTs integrate multiple digital twins with comprehensive data and knowledge, spanning the entire lifecycle of a system. By leveraging semantic technologies, CDTs offer superior comprehension, decision-making capabilities, and predictive accuracy, driving optimization, troubleshooting, and innovation efforts. This sophisticated integration results in more nuanced, comprehensive representations of complex systems, significantly boosting performance and operational efficiency within the framework of Industry 4.0 initiatives [1], [2].

Expanding the application of digital twins across various sectors, including manufacturing, healthcare, petroleum, traffic management, smart cities, aeronautics, and agriculture, highlights the technology's versatility [3], [4]. However, it's important to note that beyond manufacturing, the use of digital twins is often at an early stage, focusing primarily on the development of 3D models rather than on advanced analytics, artificial intelligence, and comprehensive integration. The manufacturing industry, in contrast, demonstrates significant maturity in digital twin technology through the deployment of operational solutions, showcasing the potential for growth in other sectors by adopting data-driven decision-making and predictive capabilities [5], [6].

A cognitive digital twin of a country extends this concept by replicating a nation's physical and social characteristics to simulate and forecast its behavior under various circumstances. It utilizes historical data and real-time information, along with advanced technologies such as artificial intelligence, complex systems modeling, machine learning, and big data analytics, to create a precise and realistic model of the country. This model is continuously updated with real-time data, serving as a tool for testing multiple scenarios and predicting the country's reaction. Such a cognitive digital twin aids policy makers in improving the nation's overall well-being and the welfare of its society, providing crucial disaster preparedness and response capabilities, and identifying potential risk or instability areas. This application exemplifies the breadth and potential of digital twin technologies to transcend traditional sectors, offering strategic insights and operational efficiencies on a national scale [7], [8].

B. LLM Agents

According to Franklin and Graesser, an autonomous agent is an integrated system within a particular environment, capable of sensing its surroundings and autonomously making decisions based on those observations. This system consistently executes actions driven by its own set of goals, thereby influencing future events within its environment through its interactions [9].

Building on this concept, LLM agents have demonstrated remarkable abilities in language comprehension, intent detection, reasoning, memory, and even empathy. These capabilities render them highly effective in tasks requiring decision-making, planning, and executing coherent sequences of action. LLMs' adaptability in tackling new tasks, adjusting to various languages, cultures, and domains, significantly reduces the complexity of training processes and data collection. This adaptability is further showcased in their ability to generate representative trajectories for task demonstrations, thus enhancing and streamlining the training paradigm for AI agents [10].

Despite these strengths, LLMs encounter limitations in processing inputs from diverse modalities, grasping implicit relationships within their environments, and acquiring essential world knowledge for advanced intelligence development. The challenge continues in enabling LLMs to efficiently plan and execute actions based on their understanding. Furthermore, ensuring adversarial robustness, trustworthiness, and mitigating potential risks such as misuse, unemployment concerns, and existential threats to humanity pose significant challenges. These issues highlight the intricate process of developing LLM-based agents that are not only effective and efficient but also safe and ethical in their operation and interaction across various contexts and societies [10].

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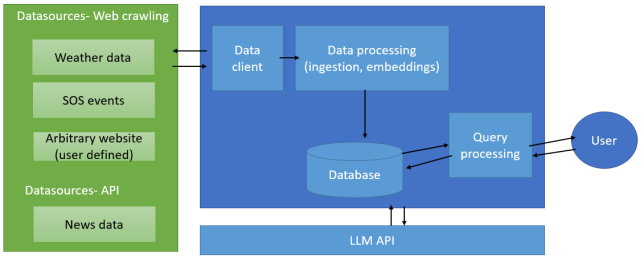


Fig. 1. Conceptual design of agent architecture

In [11] authors present a framework that integrates LLMs with digital twin technology and modular automation systems for dynamic and intelligent production planning and control. The key contributions include the creation of a scalable, intelligent automation system capable of interpreting and executing complex tasks using LLM-agents, and the development of a structured approach for injecting task-specific knowledge into these agents, paving the way for more agile and adaptable manufacturing processes.

III. Methodology

In this section, we outline the methodology employed to construct the LLM agent arhitecture.

A. Prompt Engineering

Prompt engineering constitutes the process of directing LLMs through detailed instructions or prompts, which establish operational parameters, including context, regulations, and targeted results. By tailoring these prompts, users can fine-tune the LLM's functionality, steering output generation to comply with specific criteria like coding conventions, programming approaches, or detailed analysis of designated keywords. This technique amplifies the effectiveness of LLMs across diverse software engineering endeavors by yielding more organized and pertinent results [12].

B. Web Crawling

Web crawling, also known as web scraping, involves the automatic extraction of data from websites using software tools. It is used to collect and compile data from vast numbers of web pages for analysis and utilization in various applications, particularly in fields like business intelligence. This process enables the gathering of valuable information that may not be readily accessible through traditional means, making it an essential technique in the modern digital landscape [13].

IV. Conceptual Arhitecture of Agent

This section presents a comprehensive overview of the agent's architecture, as illustrated in Figure 1.

A. Data Description

The idea is that the architecture supports both textual and time series data. Data sources can either be predefined, fetched from a specific space, or obtained on the fly. Predefined sources include essential and informational data, such as SOS alerts, news, and weather updates. Meanwhile, dynamic sources, typically websites specified by users, offer detailed information on particular issues like traffic congestion. They incorporate real-time data on traffic density, roadworks, and other relevant details. This structured approach ensures the digital twin remains dynamic, responsive, and a comprehensive representation of the country.

First, data is collected from available APIs, representing the most structured form of data input. The digital twin accesses various APIs to gather real-time or regularly updated information on weather conditions, traffic flow, public transport operations, environmental sensors, economic indicators, and more.

Second, data is collected by crawling web pages. Since not all relevant data is accessible through APIs, the digital twin uses web-crawling techniques to gather data in various formats, such as HTML content, published reports, news articles, or statistical data embedded in web pages. Data extraction and processing techniques convert this unstructured or semi-structured data into a format suitable for integration into the digital twin's model.

Lastly, data and descriptions are provided by individuals. This includes data descriptions, expert insights, local knowledge, and other qualitative data not captured through APIs or web crawling. Analysts and domain experts contribute descriptions, feedback, and corrections, enriching the digital twin's performance.

B. Data Ingestion

The architecture employs an LLM to create SQL scripts that add data into a database. For each data source, there are separate CSV files stored. The system then examines each data source's file to understand its format, including the data types and column names. Using this information, it asks the LLM first to generate a script to create the table if it doesn't already exist, and then to create an insertion script. This insertion script is used to add data from the CSV files, which are automatically extracted from the digital twin system.

C. Agent Workflow

The process begins with the user's prompt and moves through several steps performed by the agent, utilizing predefined prompts.

In the initial step (Figure 2), query enrichment is performed using the prompt, as shown in Figure 3. This process integrates additional, contextually relevant data with the original user query to refine the system's response. This enhancement allows the system to deliver a more

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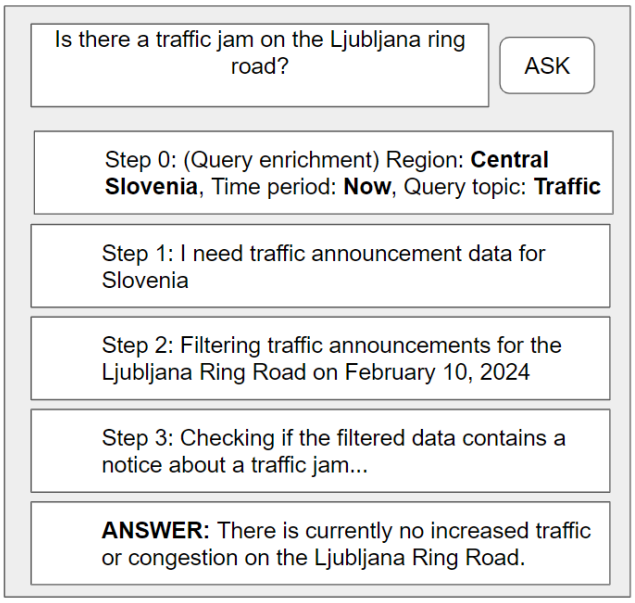


Fig. 2. A step-by-step abstraction of an agent's process, from receiving an initial question to synthesizing and presenting the final answer, involving multiple intermediary actions

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Prompt: Given the query "Is there a traffic jam on the Ljubljana Ring Road?",
enrich it with additional information, such as region of interest and time of
interest.

The following are the examples of successful enrichment:

1.  Query: "Is there a traffic jam on the Koper Ring Road?"
    Region: Primorska
    Time: Today
2.  Query: "Was the Vršič Pass closed yesterday?"
    Region: S. Primorska
    Time: Yesterday
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Fig. 3. Prompt in the initial step enriches the original prompt

precise understanding of the query by incorporating region-specific and temporal data.

In the first step (Figure 2), identification of data sources is performed: This step involves selecting data sources using a prompt, as shown in Figure 4. In cases of unclear or insufficient information, the agent seeks further details or requests relevant web links to enhance its understanding of the context. Additionally, the agent might display a selection of database tables, encouraging user involvement to accurately pinpoint the data source required for the query. This method ensures the selection of data that is precisely aligned with the query's specific needs.

In the second step, as depicted in Figure 2, the agent carries out data retrieval and aggregation, employing the prompt illustrated in Figure 5. This process involves the system extracting information from chosen sources and applying time and location filters to identify relevant data. This data is then organized into a format that facilitates analysis and insight generation. Upon pinpointing the necessary datasets, the system extracts and refines this information. It employs time-based and content-specific filters to manage the volume of data within the constraints of LLM processing capabilities. While alternative methods

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Prompt: Given the query "Is there a traffic jam on the Ljubljana Ring Road?"
with additional information "..." and the list of available data sources
together with their descriptions:

  - Data source: Promet.si. Description: Contains the list of traffic reports
    for Slovenia for each given day.
  - Data source: ARSO. Description: Contains weather information for Slovenia.
  - Data source: AMZS. Description: Contains real-time information about the
    current traffic flow on every major Slovenian road.
  - Data source: Region Info. Description: Contains the mapping from cities to
    Slovenian regions.

List the names of the selected data sources. Additionally, list if there are
any other sources that are necessary to answer the query but are not listed
above.
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Fig. 4. Prompt in the first step, helps to select relevant data sources

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Prompt [Structured data source1]: Given the query "Is there a traffic jam on
the Ljubljana Ring Road?", additional available information "Region: Central
Slovenia, Time: Today" and the data source "Promet.si" with the description
"Contains the list of traffic reports for Slovenia for each given day",
formulate a query following with the following fields:

  - Region: traffic information about which region should be retrieved.
  - Time: traffic information for which date should be retrieved.
  - Report type: …
```

Fig. 5. In the second step prompt facilitate querying over relevant data with time and location filters, enabling aggregation

like text2sql are explored, they often encounter issues related to language compatibility and the precise understanding of semantics, especially when processing data in languages other than that of the query or dealing with synonyms and variations in terminology.

In the final step is generated response, (Figure 2) is where synthesized dataset is processed through the LLM (Open AI Code Intepreter), enabling the agent to formulate a comprehensive response to the user's query. The response not only addresses the query but also includes references to the original data sources. This aspect of referencing is crucial as it offers insights into the logic and basis of the response, enhancing the user's understanding and trust in the information provided. Moreover, it mitigates the risk of reliance on potentially inaccurate or 'hallucinated' answers, ensuring a higher degree of reliability and transparency in the system's outputs.

V. Conclusion

In conclusion, this paper has presented an architecture for enhancing interactions with cognitive digital twins via an LLM agent, focusing on dynamic interactions. This architecture is built on the foundation of natural language processing and prompt engineering, enabling the system to handle data aggregation from diverse sources and facilitate analytical capabilities through conversational interfaces.

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Prompt: You are given the query "..." and the contexts retrieved from multiple
sources:

  - Source: Promet.si, Source description: …, Retrieved context: "..."

Analyze the context and construct the final answer. You can also use the code
interface or other tools:

Provide the answer in natural language.
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Fig. 6. The prompt in the final step assists in analyzing a query within its context, retrieved from multiple sources, to construct a final answer in natural language

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The methodology section detailed the construction of this architecture, highlighting key components such as prompt engineering, web crawling, data description, data ingestion, and query processing. It described how the agent integrates and processes data from APIs, web pages, and individual contributions, streamlining the inclusion of new data sources and ensuring efficient, accurate query responses. By leveraging a structured process that enriches queries with contextual data, selects relevant sources, retrieves and aggregates data, and analyzes it to produce informed responses, the proposed system offers a scalable, adaptable solution. This enhances the comprehensibility and management of extensive datasets, making it a significant step forward in employing cognitive digital twins for sophisticated decision-making. The integration of user feedback mechanisms further allows for the incremental refinement of the digital twin, potentially augmenting its value for decision support. Overall, this architecture represents a comprehensive approach to leveraging LLM agents for more intuitive, efficient interactions with cognitive digital twins, promising enhanced decision-making capabilities in complex environments.

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